

Karl Friston ~ Active Inference Insights 001 ~ Free Energy, Time, Consciousness

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excellent and we're away okay so welcome everybody to the first episode of active inference insights brought to you by the active inference Institute I am your host darus parvey Wayne and today I have the privilege of speaking to Professor Carl friston one of the most revolutionary thinkers in the fields of Neuroscience cognitive science theoretical biology and dare I say physics and philosophy a leading Authority in neuroimaging he is the inventor of statistical parametric modeling and voxel based morry I hope I pronounce that correctly more recently he has posited the free energy principle which is technically a normative account of self-organization in terms of optimal Basi and design or in other words a description of how things continue to be things as such his groundbreaking ideas hold profound implications not only for biology and psychology but also the the very nature of reality itself ladies and gentlemen it is my honor to introduce Professor K friston Professor friston thank you so much for joining us well thank you thank you for that lovely introduction what you can say is I I just invented lots of three-letter acronyms that's much simpler so you have and I sense a lot of freeletter acronyms have also spawned from from a lot of your work um excellent well this podcast this is the first episode so as I was saying sort of off camera this is very exciting for me hopefully very exciting for everyone tuning in the kind of uh the reason why it's come about is that we wanted the active inference Institute wanted a way to introduce the ideas of active inference and the free energy principle perhaps to an audience who were not PhD holders scholars in the area but had a kind of latent interest in Psychology Neuroscience maths biology whatever it is and wanted to explore these ideas a little bit fter a little bit further so it's a non specialized introduction to the ideas of active inference and the free energy principle and as the author inventor discoverer of the free energy principle we can have an interesting philosophical conversation about what that is um it's it's an absolute Delight to have you so I guess my first question that I need to ask for everyone listening is could you give us a very brief or not very brief but a very um accessible definition of the free energy principle um what

free energy is and how you know what this what this principle says about reality and ourselves yes I'm not very good at giving brief answers but do my we have we have a while so so go ahead um so just to put things in context um the free energy principle is a principle um as read by a physicist or engineer that means it's a method um much like Hamilton's principle of least action that you can apply um so one might ask what are the S the most important applications of the free energy principle I guess that would be active inference so I will answer your question and I will say what the free energy principle is but um that may have to be qualified by looking specifically at the application which is active inference and what that means um theologically her istically you know what it brings to the table when you do apply the free energy principle but the free energy principle itself um is just a description of things that persist or exist and it rests upon a careful specification of what you mean by a thing um it starts with the exactly the same assumptions that all of physics starts with basically a random dynamical system some people describe that in terms of um stochastic differential equations for if you're not familiar with that kind of description is just saying that you can describe the universe in terms of the way that states change as a function of themselves so it's a very generic um description of Any Given universe that just commits to the notion that there is a state of the universe and it could be very high dimensional and that it has some Dynamics so that's where the free energy principle starts but the special thing about the free energy principle um that distinguishes it from other um formulations in physics for example Quantum thermodynamics um statistical and classical mechanics all of which can be if you like derived from this basic description of the world as a random dynamical system the thing that um separates the free energy principle uh or gives it its particular flavor is a careful distinction between the states of something and everything else uh and the way that we do that is by inducing a separation you know uh technically a partition of all possible States into three kinds the states of something that are internal to the thing everything else external to the thing and then crucially a set of states that separate state the uh internal from the external states that allow and couple the inside to the outside and that's referred to as a um the blanket States or a Markoff blanket or boundary um so a lot of this rests upon this this notion of a boundary that um on the one hand separates the inside from the outside of something and by a thing I mean anything from a small particle to a to a priest or a person anything that can can be individuated and uh possesses characteristic States constitutes something um and with tongue and cheek you therefore have a theory of every space thing not everything but every thing um so this Mark of boundary or Mark of blanket um um is a sort of crucial device um in the sense that it

allows you to individuate something from everything else but at the same time it couples the internal to the external States so we're now talking about a special kind of physics and probably a more ubiquitous kind of physics which is the physics of open systems because the inside is now vicariously open to in exchange with the outside across the Markoff blanket and just for interest because this will be relevant later on for active inference we usually divide the blanket States into active and sensory states that can be regarded very simply as of the inputs and the outputs um technically the inputs of the sensory states are defined in the sense that the internal states don't influence the sensory States and conversely the active states are defined such that the external states don't influence the active States so but more simply what we're saying is that something can be defined in terms of the influences upon it and the influences that it exerts through the blanket states on the outside world so just given that assumption that entails existence in the sense that to Define this Markoff blanket you have to you need for the system to have some characteristic States technically an attracting set of states that you'd find the thing in just by assuming the existence of this attracting set of this random dynamical system that possesses this distinction between the inside and the outside everything else follows and everything else simply put is you can read the internal and active States sometimes refer to as the autonomous states of a particle or a priest or a person um as either if you're a physicist conforming to a particular principle of least action and that principle of least action is that it minimizes the path integral of the variation of free energy um if you want to think of this more intuitively because that free energy also stands in for or is an approximation to a bound upon model evidence you can also say that it must be the case that the autonomous states of anything will look as if they are trying to maximize the evidence for what we would now consider the internal states to be um a model of what's going on on the outside so some people refer to that as self-evidencing um so if I just repeat that from the physicist point of view given the assumption that you've got this random dynamical system with an attracting set that has this partition that allows you to individuate something within this system it has to be the case that the Dynamics of the autonomous states of that system um perform a gradient flow on a quantity called variation free energy that's it is that interesting well it becomes interesting if you realize that this variation free energy has a particular meaning and a particular interpretation that gives you a teleology for this kind of dynamic uh and that Tey is often referred to in terms of bayesian mechanics why well because the free energy the variational free energy is the same quantity that measures the evidence for a generative model of the

causes of some data if you're a statistician and in this instance the data that we're talking about are the impressions of the outside State on the on the Markoff blanket so um what you're now in a position or you now in a position you are now licensed to talk about anything that exists in some simple Elemental sense uh performing inference and possessing this or evincing this kind of basing mechanics um and in philosophy um people like yakob har nice summarizes as self- evidencing so you can imagine that it looks as if anything that exists is basically moving in a way uh sensing and acting upon its World in a way that's Garing um evidence for its own existence so it's this is you know if you like a slight cheat in the sense that you what we're doing is describing the Dynamics of something that exists and then we're saying ah but it now looks as if it in in in virtue of its existence it is now it can be described theologically as acting to gather evidence for its own existence its own um um its own structure and that structure entails a generative model of what's going on on the outside so how does that help well it doesn't really help in any fundamental way other than it's very pretty and it it's it's nice to talk about basy mechanics as being a sort of complement uh or another kind of physics that comes from the same stable as quantum mechanics and cystic mechanics and I repeat classical or Iran mechanics um however there is a nice um application at hand once you realize that this uh once you interpret um the variation free energy in terms of a um technically the logarithm of some model evidence what that means is that you can now build things you can build things and you can simulate those things why well because you know that their Dynamics must be um conforming to this principle of least action um uh another way of thinking about this is that it um does a gradi in descent or hill climbing on a particular um functional which is um this um variation free energy or the negative variation free energy which is the log evidence so um because this log evidence uh is a function of a model that just is a description of the characteristic States things occupy or things are found in um you can now write down the generative model which is just a probability distribution over the characteristic States or defining the characteristic States you want your thing to be attracted to this is the attracting set and once you've written down your generative model you can then work out the gradients um of the free energy functional of that generative model and then you can simply um integrate or mathematically solve for the Dynamics the flow the motion the movement of this thing so I know this isn't terribly accessible but it's it's uh uh a way I think of these things that you are uh now in a position instead of just looking at something and saying oh I wonder what uh generative model um would be fit for purpose to describe this particular thing you can now turn that on its head and say okay I want this kind of thing and I'm going to describe and

Define this kind of thing with a particular generative model I just have to write down the probability distribution over the causes and consequences from the point of view of the thing in question namely the external States and its Impressions upon them the Markoff blanket and once I've written that down I can now simulate the autonomous States I can now create a little autonomous artifact a little autonomous particle that behaves in exactly the same way as something that existed that had that kind of generative model so it's a very practical if you like um uh it's a principle that can be applied in a very practical way to simulate things and if you can then simulate at things you can start to build things you can start to um um do computational Neuroscience you can start to sort of naturalize cognitive um neuroscience and possibly even sort of um psychology in terms of the physics that is inherent in this basing mechanics that comes from the free energy principle um you can also start to ask questions about the beliefs of another person why well part of this generative model or um the generative model can always be sort of um split into two things sort of priors before you see some data and the likelihood the likelihood of that data given um the causes of that data these will be the external States so you have prior belief over the causes the external States and you have a likelihood that Maps the causes to the consequences which are your observations so your Sensations um and therefore you can now fiddle with and say um a synthetic subject or person who is being simulated in the computer until their behavior their actions possibly their decisions matches that of Any Given um cohort or any given um um subject or animal um and then having adjusted the prior part of the generative model and possibly the likelihood you've now basically created a digital twin that emulates the same kind of belief updating the same kind of basing mechanics um that you can now infer as possessed by the thing in question so this is a um sort of quite a high-end application which which is um you know one of the the um the motivations for um de for developing the free energy principle and and active influences an application um that it enables you to essentially phenotype another thing another person for example in terms of this belief updating in terms of this basing mechanics um and that can be quite useful in context when you want to understand how people work and particularly how their belief updating works this becomes particularly interesting in the context of computational Psychiatry where you want to be able to explain people's behavior in terms of what do they Bel about the world out there you know how how um how are they sense making how are they assimilating evidence for their world model which you can read as the generative model U and under what priors are they doing doing this sort of basian um assimilation or evidence accumulation um the underwrites their decisions so that's one um um

one particular application there be other applications um you know if you want to make intelligent artifacts or you want to understand the mechanisms of belief updating at different temporal scales um belief sharing between two things that are in communication or transacting across the a shared markov blanket or indeed uh looking at a shared World um and then one can start to get into the world of um distributed cognition and Federated inference there are all sorts of things you can do once you start to apply active inference yeah that was that was fantastic I mean we have a lot to get our get our teeth into um and and probably some Concepts there that are new to a to to our audience so it might be worth sort of backtracking certain things and and and trying to um find a kind of yeah a let's sort of dissect some of these Concepts together um the first thing I want to speak about is this notion that the internal States in some sense embody um the kind of desirable model given their attractor State the desirable model of the world I think um one of these one of the questions that one has when one enters into the world of active inference is whether an organism is a model or whether an organism has model and here we sort of have a a backtrack to cybernetics and the good regulator theorem that all good um regulatory systems have a model of their environment if we think about thermostats or um anything alike so maybe it's worth un unpacking that because I think a lot of people might have this intuitive sense that there's a homunculus in the brain which has a model of the world and it's very internalist but maybe from a more physics perspective we can adopt a more externalist position position where there isn't this little thing which has a sort of Representative picture of the world but actually is just enacting um some model of itself which in many ways is just some it some version of the of its external Dynamics um so let's start there because I think that will feed nicely into maybe a conversation about internal and external Dynamics and exactly what that looks like in the free energy principle yeah that that's a great question and uh you some sort of deep issues Unearthed by that and it's nice you've introduced cybernetics and notion of the good regulator theorem because you know in one on one view I think you could look at um the free energy Principle as just a 21st century version of that kind of cytic thinking there are other there are other things that the free energy principal inherits uh from um you know James's Max men principle lots of ideas in in in Psychology perceptions hypothesis testing there are lots of different um roads that converge upon this um um formulation um that that is uner in terms of the free energy principle um but that particular question is is really interesting uh I have to say that I usually um try to elude that question by saying that the the internal States entail a generative model so um however I um if I had if one was pressed upon it um then um I would say very clearly uh that the internal States um the

thing is a generative model it's not if you like um a picture that you painted in terms of that hunus looking at its sort of you know its virtual reality inside its head and trying to what's going on it is instantiated it is realized in its physics in its existence in its Dynamics um the reason I use the word entailed is that the generative model um is um technically uh just a probability distribution um so you you ask yourself well does a probability distribution exist um in some physical sense um and the answer is well no um a probability distribution is what it is it's a a mathematical description um however that um probability distribution will have sufficient statistics that could exist so what I'm saying is that um for example let's take a ging distribution um you know I can I can um Express mathematically I can write down the notation for your ging a bell-shaped distribution over some variable that goes from plus to minus infinity um or I can say I can characterize or specify this ging distribution in terms of its mean and its variance and and those are two numbers that are not random variables they are sufficient statistics and it may be and in fact it is the case that that the um the free energy principle Associates the the physical internal states with the sufficient statistics of these basian beliefs and there's a lovely story you know in mathematics um that would take you down the path of information geometries and cisal manifolds and um belief updatings as movements in these geometries and you know try to down the the metrics in these spaces and how that relates to uncertainty and precision all all that's a lovely story um we don't need to really pursue that at the moment the key point being made here that the generative model um is not physically instantiated other than in terms of some uh sufficient statistics in any setting um however in the in the free energy principle the generative model is in of itself never actually instantiated in terms or realized in terms of um its um statist sufficient statistics and that's because of something that we were talking about before which is the Dynamics so the Dynamics is a gradient flow um which means that all we need are the gradients of the free energy so we don't need the generative model all we need is the gradients of this free energy function or F technically a function or a function of a function of the generative model so if I was building an artifact I could certainly prescribe the right Dynamics by writing down a generative model and then I could evaluate the gradients of that um of that um free energy function of the generative model and then those gradients would actually drive the Dynamics I could simulate self-organization I could simulate active inference I could simulate belief sharing whatever I wanted to do um but in that simulation all I really needed were the gradients of the free energy uh and we only need the generative model because the um the free energy is a function of the generative model so in that sense the generative model is really a t

logical description that you would bring to the table to understand your sense making of another thing um the thing in of itself just needs the free energy gradients it just needs to self-evidence by moving in the direction of maximizing the on average the log evidence for some abstraction which is a generative model um but it doesn't need to know that um um just to you know so we don't confuse people so moving up those log evidence uh gradients very much like um you know paradoxically moving up uh concentration gradients to Clump together to keep in this attracting set and to resist the dispersion of the random fluctuations inherent in these random dynamical systems is just the same as moving down the negative log evidence which is the um um the uh the free energy itself it's it's it's it's the same thing so what does that mean well it um what it means is the joint model is something that you bring to the table to um endow the behavior of this thing with an explainable teleology um of the kind it looks as if this thing is acting in a way to maximize the evidence or gather the evidence for its generative model and this is what I think its generative model is but notice the thing in of itself doesn't go through this process um it could of course so if I now people things like you and me we we do have a model of mod we do have um and we communicate it and we talk about it but most things you know um um um uh most things would not would not possess that um that deep structure so you can invoke hular you can invoke self-modeling you can invoke um um your uh the kind of um not hermanus in in the sense that you would normally um um end up with an um you know with with an infinite regress but you know the the kind of metacognitive aspect that comes from any deep or deeply structured hierarchical generative model you you you you can you can get that sort of humulus like Detachment and metacognitive uh perspective on things but strictly speaking um you know the generative model is is is only entailed by the den DCS and I should say that you know this is probably best to think about um you know applications of the free energy principle in you know in in um as being limited to simulating and modeling and understanding other things as opposed to oneself unless one's doing a lot of self modeling um so so in that sense you are ascribing um a teleology a purpose um um that would explain the behavior of something that you're observing but notice because you are observing it from its point of view you are an external State and from your point of view all you can see is the Markoff blanket you cannot see what's going on on the inside so it is um it would be mathematically impossible for you to ever know whether what's going on on the inside is indeed um describable as even a gradient flow on a free energy functional um let alone did it have its own internal model did it had it have its own beliefs it just looks like that from your point of view because you'll never ever know by definition if you knew then you

would um by definition have breached the Markoff blanket and that thing would cease to exist as the kind of thing that it was before um I mean may sound obvious or it may sound trivial um but I don't think it is if you just look at people like me who spend our entire lives trying to peer through the Markoff blanket using brain Imaging for example and you know or people doing um comparative Anatomy by dissecting dead things um or by inventing new techniques to try to try and get uh you can never get beneath the Markoff blanket but you can certainly sort of um have certain perspectives on the Markoff blanket that are closer and closer to the internal States but you can never actually get in there I mean you know there's I imagine a whole philosophy on this that you you you couldn't um see your own um visual processing you couldn't hear your own cellular Dynamics um you you you you can't um ever have access to the inside very much in the spirit that internally would say you can never have direct access to the outside um so I think that also applies to these stories about generative models and the basic mechanistic interpretation of uh that comes along with the uh with the free energy principle does that help in some way it's a little bit subtle and I can just imagine a lot of my friends tearing their hair out because they have very particular views about this yeah some people think it's it's a model of a model some people think it's you're taking very internalist some people external mathematically um you know it's it's not the model which drives drives as you know it is the it is the Dynamics which drives us and that Dynamics can be interpreted in relation to a generative model should you want to excellent well I'm gonna follow that philosophical line of thinking and what I want to ask is this is an interesting point that the notion that we can get to a closer approximation of what the internal States of ourselves are for example but it's always going to be an approximation because piercing the Markoff blanket would constitute self disintegration under the uh rules of or the principle the free energy principle per se how how in a sense therefore do we know or or how how how uh how much can we even make the claim therefore that there is such a thing as an external Dynamic that goes beyond anything goes beyond our sensory inputs um it may seem like a deeply non-scientific question but um you the way you were speaking couldn't I couldn't help but think of people like George Barkley and even Canton this phenomenal Nal distinction if we don't ever have direct access to so-called external Dynamics because that is part and parcel of the the physical game that we're in why can't we just say that there are no hidden States and all we get are probability distributions over sensory appearances but we don't need something like a likelihood mapping to something that's latent um well you you you you could say that and a lot of my friends do say that um and I think in the sense that um the the notion of direct access

means that um you will never know in some sort of fistic sense um what the external states are if they exist at all and and um um on the other hand you could also argue um that um there is profound access to the um and existentially meaningful access to the external States it's just organized in such a way that it it has to be transacted by the Markoff blank blanket um so you know I haven't I haven't this conversation for for for several years now so I used to have lots of Pat answers which um depending on who who I was talking to but the the you the way that you phrase a question which was very clever um this made me think well you know what is the internalist argument does it mean that because I don't have direct access that the external states don't directly influence me that means that they don't exist um is is you know is that the the argument that an internalist would make well may might it be the case that all we have we we don't need to invoke a kind of further realm a further causal realm an underlying ontological realm if all we have are just the streams of sensory input and seemingly some regularities some patterned regularities in them why at least in terms of doing the basian modeling do we need to invoke a hidden State an underlying cause for examp like if we just take that if look at the low road TCT of inflence we talk about how human we talk about how cognitive creatures like ourselves predict the world let's say at a very simple level I a question I've always had is why do we why does the claims to predictive processing or predictive coding more generally why does it make claims about hidden States why doesn't it make claims just about regularities within streams of appearances that's the way that I would pose that question right um so I'm just mindful you've introduced uh predictive coding and predictive processing we have we perhaps it would just just to for those people um who may not uh know that the the intimate relationship between all these things so another way of looking at free energy is it's the uh amount of prediction error and more specifically if you're talking to some people it would be the uh the total amount a Precision weighted prediction error um which means that there's another reading of minimizing free energy which is not the maximizing the model evidence but um minimizing surprise where surprise is the implausibility of some sensory data um given your model of that data so in exactly the same way that if I go around Gathering data that I find uh continuously surprising um then that's basically evidence I haven't got the right kind of model to be um to be able to explain I.E predict those those data um so again we have one of these um um sort of completely equivalent descriptions but you know that have very different uh flavors that to maximize model evidence is to minimize surprise which is to minimize a prediction error which is to maximize predictability which is to minimize very free energy and so on and so forth they're

all the same they're all the same thing so when you're talking about productive coding as a particular um algorithm for minimizing free energy uh technically um you know a variational scheme which is known in engineer as a Kalman filter um what you're saying is I'm making certain assumptions um about my generative model um that means I can write down these free energy gradients as prediction errors so literally the prediction error my my Po the Precision weighted prediction error literally is mathematically exactly the gradient that we were talking about before um so just join the dots um so to come back to your question um uh why is it that um I need to invoke latent States in order to make sense of data um I would argue um that the sense making is just explaining data in terms of Latent States um and those latent states are often referred to in um you know in the technical literature as hidden States and what do you mean by that well it just means they're unobservable from the point of view of the free energy principle they are hidden behind the mark of blanket so I would say that detecting patterns simply is invoking latent states that organize and structure those patterns um so um I don't think there's anything magical about about latent States so the latent states that are if you like entertained by your internal Dynamics and your sense making um uh are not the external States you they are descriptions that are only relevant to the generative model that um you need in order to define the prediction errors or the free energy gradients in order to um you know simulate or describe um your sense making in terms of your neuronal Dynamics or your internal Dynamics or the operations of say a thermostat but there's a Twist here um there's a reason it's called active inference because it's got a big active at the front um so it's not good enough just to say oh I can understand psychology just as making lots of sense of my sensory Impressions that's that's not good enough um because you know you are um coupled to the world um in in in reciprocally which means that you actually um are not only are you exposed to the world but the world is exposed to you in an exactly symmetrical way and what would that um you know how could one describe that and then we come back to where where we started with this partition of blanket States into sensory and active States into the inputs and the outputs um which means that sense making is quintessentially active so one way of phrasing that is that yes I'm making sense of all this sensory data and all the patterns um but at the same time I'm actually in charge of and actively soliciting those data causing those patterns um so I think once you um once you bring that sort of inactive um aspect to the table so notice um I Notice that you you you treated predictive coding as a as a sort of generalization predictive process in it might be easier if you did the other way around because then you have the grace to um accommodate action in predictive processing its most General sense and if you don't

then predictive processing you know I think is an incomplete story you have to have um as part of the processing the active solicitation the um the garnering and the structuring and quering the world in a way that reciprocates with the right kind of information that allows you um to do the good sense making in terms of um you know the latent states that we that we're talking about um I mean forly I this takes us um into another part of the story we didn't previously unpack about active inference so somebody might ask well Okay so we've got this um um application of a um a free energy principle that a physicist might be quite comfortable with um to um behavior that entails some kind of sense making um cheekily with you know I like to call this sensing Behavior then I get told off for that um but um so uh so how does this differ from um reinforcement learning you know how how how you behavioral psychology um it differs in a quite fundamental way U because the big thing that active inference brings to the table is is that um first of all it says you are in charge you are actively uh sensing you say this is Active Vision active perception active inference in this most General sense um which means now what are the imperatives for generating your own data now you could if you were doing behavior of psychology you of of a behaviorist sort in the 20th century you could say well I just want to generate those Sensations that I found I'm rewarding um and so I got some um privileged sensory um channels that um I'm going to label as as reward and I'm just going to act in a way that maximizes the reward that comes in from principle that's not what that's not what happens what happens is I want to find those data that resolve the expected surprise or the expected free energy consequent on acting soliciting those data so what's expected surprise well technically it's an entropy it's just a measure a mathematical measure of uncertainty so what that means is that the um the the description of things that persist or exist over some particular period of time um reduces now to a kind of basing mechanics in which the behavior will look as if it is information seeking and uncertainty resolving um it will look as if it is responding to these epistemic affordances under constraints and it is those constraints that um um if you like would be the homologue of the reward um they are literally the constraints endowed by the prize on The generative model the kind of um sensory states that that constitute this attracting set you know to which I am attracted but the the vast majority um of the drives for good um good behavior or sort of behavior that things that exist would evince um is this resolution of uncertainty this information seeking and indeed you know um that leads you into considerations well what what is the how would you describe that kind of information that optimal uh information seeking behavior um if you were a scientist or um if you were an engineer and if you were a scientist you'd be looking at the principle of optimal

basium design um articulated by people like Dennis Lindley in the middle of the last century um where you measure the quality of an action in terms of the Information Gain afforded by the data that you secure by acting in this way exactly the same idea um um emerged in the context of active learning in machine learning you know this is the problem if getting data is costly what data point would I need to resolve the greatest uncertainty about my um beliefs about the latent causes of those data um so Active Learning again notice actives uh your actives um your um an explicit part of this process so in answer to your question about you know would it be sufficient just to understand sentience in its most basic or Elemental form as discovering patterns in data I would say no um because that denies the openness of Any Given system immersed in her World um in two directions you know the world influencing you and you influencing the world and since you put that into the mix I think it's very difficult to conceive of any construct um possibly not in in philosophy um but certainly any construct in computer science or um um uh or physics that would um um that would permit a complete description of senting behavior just in terms of sensing patterns does that make sense yeah yeah that makes sense that makes sense um makes sense to me um yes I want to stay on action for a second because I think the the notion of uh the minimization of free energy or the maximization of mod 11 makes some uh could make some intuitive sense for an agent like ourselves in the sense that we we have our sensory inputs which are Divergent from our prior and we act to change the world to make those those sensory outcomes more in line with our expectations our preferences however at a more fundamental level uh one one definition that we spoke about the free energy principle is that the internal and external States look like they're tracking one another across a mark of blanket now a question that comes to my mind when when I see a definition like that which is I that makes sense for me uh for a sentient conscious human but what would a stone be tracking or what would a drop of oil be tracking to use an example that you've um used before I think so using going to something a bit more less animate and I presume the answer lies in animacy um I think would be a really interesting Avenue to go down because it gets these fundamental questions of how does something as minimal as a particle retain its partiel and and and what does that really look like when when we don't when we presume that they're not conscious unless we're pan pists but we won't go just down that route just yet well that's where you were taking us which which is a fun place to be um right how does how does one um elude pan psychism in the free energy principle I well I think you you've just you've just uh answered your own question there but so perhaps we just unpack that a little bit um yeah so what's the difference between a stone um and a thermostat

and and and me um and I think what you're what you're now asking is um are there um what I will call Natural kinds but I wasn't allowed to because that has a lot of philosophical baggage but for the purpose of this conversation I'll just say are the natural kinds of things um that display different kinds of behavior that can that can be equipped with different kinds of teleology and I would say absolutely uh and you've just rehearsed your key so as you noted um to be animate um is a big thing um in the sense that the stone is not animate um in the sense that it moves itself you know Stones started you know wandering uphill or flying around that would be much more interesting um but they don't have that animacy that is characteristic of biotic self organization so what does that mean in a sort of um deflationary sense of the free energy it just means that the active states are an empty set um and usually um for Stones you might also argue the internal states are empty but you don't have to yeah the stone can be um you know making sense of its World in terms of its latent States or can acquire that interpretation perhaps that's a nice example of what we were trying to get um drill down on earlier on in terms of um is a generative model I mean I one could argue that a stone has a generative model of external milieu in terms of the temperature and the latent variable which is a hidden State it's a latent State you as well known in statistical thermodynamics which is a sufficient statistic of the distribution of things and the temperature uh um could well be uh inferred by the internal the interior of a stone and therefore it is making sense of the pattern of its sensory exchanges with the world on on the surface of the stone simply because it's warming and Cooling and therefore tracking the ambient environmental temperature so that's perfectly free energy principle consistent um it's a very boring kind of uh of sense making but you know and it is that boring kind of sense making that I had in mind when talking about um if you preclude the active part of active inference and just think about sort of perceptual inference um then then you're sort of missing the point really um but I think a stone is a good example of that now would you say the stone has a generative model well you could you could argue that but uh you know it's you trying to trying to um if like explain the behavior of a stone but the stone doesn't have very much behavior um but you could you could start to think well now I can test a hypothesis it is actually registering um uh the the temperature on the inside and again just to rehearse the argument the fundamental argument we were talking about before you will never know though until you break the stone because once you broke the stone it's no longer a stone it's just a broken Stone uh so you're exactly the same problems or issues confront you even with the stone but you know focusing on on that sort of big move from things with and without active States I think then you're talking about things that are

animate things that have um animacy and act upon the world um is that sufficient to get the kind of behavior that we've been talking about which is um you know the kernel of active influence in terms of um what active influence brings to the table in term in terms of uh information seeking and having a variety of different affordances particularly epistemic affordances um as an explanation for for for um for behavior um you could argue no you a thermostat can act upon the world it can switch heating elements on or off and it senses things by it by its Thermo receptors and uh you could say like very much like a wats governor um it you know it now has active States it has inputs and outputs and it has internal Dynamics and um um and in one sense the wats governor is a good regulator and therefore conforms to um the cybernetic view of of of of a generative model and it's acting in the right kind of way so does this kind of system um have that the Curiosity that would be associated with information seeking and resolving uncertainty and you would argue well no it doesn't so now you got another kind of natural kind um which does so what's the difference between you know a what's Governor or a thermostat or possibly a virus and um and me well I think the key differen is a particular kind of um self-modeling that just comes from being very big and a bit structurally complicated in the sense that there are certain parts inside my body and inside my brain that are so distant from the active states that are actually moving um my arm my actuators or um are engaging my autonomic reflexes that I cannot see them directly um and if that is the case the only way that I now can sense the consequences of my action is vicariously through the sensory States now this introduces a really interesting distinction between me and a virus it means that I now will start to treat my own action as a hidden or latent cause of my Sensations and that brings to the table okay well if now my own action is being inferred as a random variable then now is a separation between what I think I'm doing or what my active inference um um would um allow me to talk about in terms of what I what I'm planning to do or what I'm inferring to do in the spirit of planning as inference and what I'm actually doing so this um brings you to an even more a different kind of thing which would be something like me um that now has beliefs about um its own action um which is distinct from the actual real variables you know um that constitute the active states of the markof blanket um and as soon as I have beliefs about my own action I have to ask myself well as a physicist I would ask well what how what what are the probability distributions over my um my active States my um those states that um actually change the causes or the um the the external States generating my Sensations and when you write that down um what you get is exactly what we're talking about before which is this mixture of expected Information Gain basically expected free energy that

has come be decomposed into this information seeking part and the constraints afforded by your prior uh the prior part of your generative model the things that you find very surprising per just to unpack that um um intuitively you know let's take the predictive processing surprise minimization view of um self organization and um an active inference and that means that I now have the pro I am the kind of creature that exists therefore I must be minimizing my surprise furthermore I now have beliefs about my actions I have pry beliefs I must have pry beliefs about my actions of course this is all subpersonal so what kind of Prior would I have about my my own actions well I'm going to um because I exist and I'm a um a free energy surprise minimizing kind of thing because I exist then I must choose those actions that minimize the surprise I expect following that action so that means I'm going to minimize my expected surprise now there are two ways in which I can do that and we've already uh spoken explicitly about the two ways one way is just to notice that the average surprise the expected surprise um is just uncertainty so I can reduce my expected surprise or my anticipated or my um average um surprise um by um getting to grips and knowing you know what would happen if I did that uh with greater Precision so this is information seeking this is the sort of the Curiosity um it's the novelty seeking part it is the exactly the part that underwrites the um the basic the principles of optimum basian design getting the data that minimize my uncertainty because uncertainty just is expect surprise but there's another way of doing minimizing expected surprise if I know what is very surprising if you know by being um um at very very low temperatures or being very poor or being snubbed socially um everything that is if you like not characteristic of me given that I am you know I exist in you in a particular way um then any deviations of that basically uh can be thought of surprising so if I Stray Beyond very much in the spirit of um homeostasis although now we're talking about Al estasis because we're talking about the future um then I'm going to choose those actions don't just uh do the information seeking responding to the epistemic affordances but also will elude those kinds of surprising states that are very uncharacteristic of me being very poor missing out on that opportunity uh being embarrassed uh being in pain um although pain is probably a bad example we're talking about the sensations that that we want to want to avoid so um that would speak to this um um the prior um the prior cost or the prior surprise the negative which was was the preferences you you mentioned before so um you can look at these constraints afforded by the prior the pragmatic part if you like of um of these affordances some people call them instrumental so you got epistemic and instrumental um affordances that just are the expected Information Gain and the negative expected cost which would be the um your

manifestation of the constraints you could look at the complement of constraints in terms of where I don't go in my sensory State space um as um where I do go which are my preferences my preferred characteristic attract attracting sets so uh just to remind myself and and you and people who are listening still at this stage um why are we going through all this well because this is what comes out of a consideration of beliefs PR beliefs about the way active State Should unfold uh how and that becomes very pertinent when now I have to actually um embody those beliefs in my generative model so suddenly now I become something that is very distinct from a thermostat or or a virus because now I have a generative model of the consequences of my action and that generative model is very simple it just says a priori I will um a priori think that the action that maximizes information gain and maximizes this instrumental value uh are going to be the most likely policies and this you you know could be construed as a um a high-end kind of planning as inference it's not as trivial or simple as a sort of uh K control because we've got this epistemic part part part of it but it still has a spirit of planning of inference so what I'm saying quite simply is there are certain things that plan and there are certain kinds of things that don't plan a virus doesn't plan the weather doesn't plan Evolution doesn't plan the stone doesn't plan and you could argue that many um um smaller insects don't plan but as you get bigger things then they start to plan and I think you move from non-planning to planning at which point by definition you are ascribing to these bigger things like you and me um a be ascribing um a teeology to our behavior that rests upon a generative model that includes the consequence es of its own action which crucially equips it with a temporal depth because the consequences are in the future so unlike the thermostat that doesn't need to look very far into the future things that plan that are so big that they um lose contact with their actual actuators um now look as if they're they have this sort of temporal depth and this this temporal thickness to their generative models specifically enabling them to PL to uh plan in plan their actions in Into the Future so that's how I would get out of the pan psychism argument okay I've never I've never heard that argument before um about the distance from the actuators that's very interesting um because it seems in that sense that the temporal depth is Downstream on size in a sense um is that is that a correct reading and and is there a reading of your argument that you've just laid out there in which you can have deep temporal model uh modeling planning as inference and um allostasis as well as a sense of um retrospective inference without being a large thing is that or is this notion of being uh far from your actuators so that you have to start disambiguating what is your action and what is the action of the world or other people is that fundamental um yeah I mean this this is a really

interesting notion because I've I've actually never heard it really be uh elucidated like that so I'd love to know whether there is a way that a virus could be doing deep temporal modeling despite its minimal size right yeah I mean to be honest I don't know um but a a a sort of um an intuitive answer uh would be uh no size really does matter it really does matter so by size I I you know I'm implicitly talking about the um the the the sparse coupling that underwrites the conditional dependencies in manical systems that give rise to mark of blankets and blankets and blankets so um I'm not literally talking about you know how how wide something is in terms of millimeters I'm talking about the sort of um the hierarchical depth of of um condition dependences and being secluded behind markof blanket after Markoff blanket after markof blanket so um if you if you're comfortable associating um physical size with that um hierarchical depth in terms of um what a thing is in you know in terms of having this internal hierarchical structure internal markof blankets then I think the answer is very clear no you have to have you have to be sufficiently big so I'd be extremely Sur surprised um so for example I haven't said this before but you know I would imagine very very small insects can't plan whereas um something the size of a bee might be able to plan so I'd imagine that a droper couldn't plan but I I would imagine a bee might just be able to get there and it and it it starts um you know first of all that that that immediately speaks to a philosophical vagueness in terms of that temporal thickness which which which is interesting I mean is it is it categorical you know can I say this kind of artifact um can be explained in terms of a generative model that does not cover its own action because has direct access to its actuators because it's sufficiently simple and small there are no Markoff blankets that intervene between the um the you know the um effectively the the the sense making part and the um uh and for example sending predictions and the predictive processing Spirit down to the um the actuators um whereas this thing is so big that there are inevitably with probability one so many Markoff blankets that intervene there is no now direct access to action upon the world um and so that might be a uh a bright line it might be a um a qualitative distinction between things that do and do not have uh a representation of their own actions um on the other hand the counterargument I think would be this more vague notion that that you even even a thermostat could be ConEd as having a little glimpse of the future simply because it could be um cast as say a PID controller which basically um um invokes now rates of change and as soon as you invoke rates of change you've got sort of your know a mathematical image of a little trajectory into the future so it could be a great it could be a graded thing I've lost I've lost what the importance of your question what I ask you a question again it was probably it was probably unimportant um

but this is really sorry it felt I'll go again this is a very interesting line that I'd love to I'd love to dive down into so what so my interpretation of this my my reading of this is that in a sense having bundles of Markov blankets embedded in Markov blankets leads to the inevitable inference that there is something like me which is doing something and there is something that's not like me which is also causing sensory um data that I'm receiving that makes sense now let's let's tie in temporality to this does it does this does this mean that in a sense temporality is a corollary of selfhood in the sense that the selfhood is almost more fundamental than the the temporality because it comes from the dimensionality of the things that are Mark of blankets upon Mark of blankets and how can we integrate temporality and temporal thickness or depth um into this picture right yeah that's absolutely right and um you could tell the story in a number of different ways you know if you're talking to um for example Chris fields or Maxwell ramstead and they might start talking about um irreducible Markoff blankets um that necessarily uh require the notion of nested Markoff blankets and that there is some one or more core irreducible Markoff blankets that would have um this aspect of looking at one's one's own inference processes and acting internally and they get the notion of mental action and you can then start to work towards um sort of qualitative experiences in terms of self-modeling and what that might look like um the other the other way that you could tell this story is to know that um to invoke um the renormalization group if you were were we a physicist and just know that when you're talking about any hierarchical generative model there is um inevitably a a COR graining that includes time in that g in that depth um which means very simply that the deeper parts of my generative model and indeed a viruses are on a free energy principle view um um encoding beliefs about things that change more and more slowly as you go deeper and deeper so there's a separation temporal scales um so you I a common sensical view of this is that um the the neuronal representations uh down near my primary auditory cortex or indeed in in in in the brain stem um are representing very very fast fluctuations in um pressure and then frequencies and T and and sort of scenographic like um um representations that are changing um you know possibly even in fractions of of a millisecond if you're an owl um but certainly over over 100 milliseconds then I move up to sort of you know primary auditory cortex and I may be in the realm of sort of um hundreds of milliseconds and sort of frequency Glides that Define phones and I move up to secondary auditory coures and then through the auditory hierarchy getting to the level of um um words right up to um say you know some parts of prefrontal Cortex where we've gone entire semantics and syntax and narratives and stories start to emerge and as every time we go deeper

into the hierarchy we slow down or we extend the temporal Compass thereby providing a context for the faster influences and fluctuations at the level below so again we come to this um you know notion of um size entailing um a um a a certain kind of depth in this instance a hierarchical depth in the generative model that as you say necessarily goes in hand in hand with a temporal depth and a separation of temporal scales so I think you're absolutely right there has to be a temporal aspect to deep generative models and by Deep I just mean that there are nested Markoff blankets so you know to define a hierarchy is only defined in terms of the conditional independences um and implicitly the Markoff blankets of one level in the hierarchy um that defines it as a level in the hierarchy without the mark of planket the hierarchy would not be there so that's another example of this sort of um um being open but being closed um um being able to individuate something from something else and and this is as if the thing is just the level of a hierarchical a hierarchal generative model that we're ascribing to um the Dynamics of of some internal of some internal States I think there's a third story you could tell which is possibly where you're going which is um um how it relates to to Consciousness um um in the sense and this is a very simple story that that that um if I am um forming beliefs and doing um belief basing belief updating as prescribed by active inference about the consequences of my actions then that has to be in the future as we' just said and perhaps certain things think deeper into the future than than other things um but notice what we've done here it's about my actions so not only have you committed to temporality um and temporal depth and temporal thickness as an attribute of things that plan um and are more like you and me but you've also committed to agency so now you've created from something that was previously not an agent into an agent where I'm using agency in the sense that I one would not say that a thermostat had agency but you would say that I have agency um in the sense that I have beliefs about the consequences of my action and I um will or can be described as um using those beliefs to select a particular action and you know coming back to predictive coding formulations that that selection would basically I'm going to predict the state of my um Motor plant or my autonomic um nervous system and provide with the right set points and and then the the active stat will actually reflexively fulfill those prophecies those predictions um and everything is working properly I'll I'll be a good Alat or in R's terms a good hom homat um so what what you've done there is is bring agency into the game um and I think that again this speaks to the inactive part that you know you now because you've got beliefs about the future and specifically the future that you have caused that you have authored you have now become an agent so we have I think got this this stage to Consciousness and

self-awareness but I think we've got the necessary Foundation that there are only certain things that have to be agents I think to be conscious of themselves as agents as doing things um as things that exist in um um in a world uh in the sense that their existence is Manifest in terms of the consequences of the way that they act upon that world if it wasn't you wouldn't need any of this you wouldn't need this temporal thickness you just be you you just need the apparatus of a stone or a thermostat to make to do do your sense making but as soon as you got agency in the game I think then you've got you've got something which takes a step closer to um having a very deep and possibly it's this irreducible Markoff blanket um um um layer of the generative model or level of a of a deep generative model which actually now starts to entertain the hypothesis I am an agent um which you don't need but you could and you have to ask yourself why would you ever want to entertain that hypothesis I am an agent um so this would be minimal selfhood just a representation um that I can be in different states of mind when it uh that and those states of Mind are something that underwrite or I can condition on it in terms of what I'm going to do and that doing could be internal it could be how I deploy attention in terms of precision weighting in predictive coding coding um so it could be covert action or it could be overt action I could now um select um you know um certain plans at a lower level that would be then um enacted by my motor system or my autonomic system if I was that that kind of creature does that make sense it makes sense and there open up a can of worms in my head um well inevitably because we brought in lots of Concepts here and I want to try and unpick them um for people at home and for myself but it's fascinating so I'm loving it so let's keep going along this along this road we've got agency Consciousness and temporality as as these three things I want to have a little look at starting with the first thing we talk uh you you mentioned as a response to my question about temporality that we have a deep hierarchy in which um more deeper levels of the hierarchy are tracking um slower fluctuations in the external Dynamics and therefore contextualize and constrain the higher order faster Dynamics is it are we in this model are we invoking time as a fundamental aspect of the external Dynamics or is it an inference that the act that the um that the organism is making about the nature of reality that it's experiencing um well that's that's again an excellent question it's certainly um does not um um it does not address time perception uh in terms of the psychophysics of time perception um does it address um time in in a more fundamental sense I I don't think it does um in a sort of um sort of folk psychological sense um it's really um so the the answer to that question normally um is framed in terms of relative movements in a belief space um and um so if you think of time as um part of some Metric space time then

what you're talking about is time as something that can be measured time that has a distance say milliseconds or seconds on the like um could me um could me as a generative model um um ever have access to or build um hypotheses uh of time as a metric yes we certainly could um how however um it would be very difficult um you're ignoring sort of internal clocks and the like well actually no you have to ignore internal clocks you then ask how do you do it and normally what what the story the story is um it's the the number of moves you make at one level relative to another level of a deep generative model and the number of moves is quite crucial here in terms of tying that down mathematically um and my favorite way of thinking about the number of moves is basically something called the information length and the information length um refers back to something we've mentioned before which is information geometry so as if you think of our um so one picture of the free energy principle it's quite a technical picture but I think it's quite useful is that um our internal say brain States or the internal states of a thermostat um encode a probability distribution a Bas in belief a conditional belief about some latent States um which means that for every state of the brain there is if you like a point in the state space of all the physical states of the brain but each point in that space now stands in for a probability distribution and that means that um this is a special kind of space because it is equipped with a metric what is that metric well it's the K Divergence well the pathal infinites small K divergences as I move through the space so this is what this is what defines information geometry the the information geometry is a very particular um bit of mathematics um or theorizing that applies to very very special State spaces or manifolds where every point in that on that manifold corresponds to a belief or a probability distribution so now what if the metric is the um the K Divergence or you know the information length which is related to the um the accumulated K divergences as I move from one point to another um what we are saying is that the um the number of moves I make the number of units as measured with the the um this kind of metric this information geometric uh measure um which actually technically is this official information um is is um basically um a description of um the Precision by which I measure whether I've changed my mind so if we say that I've made n moves that means I've changed my mind n times and I know that because I can measure the information length and I've actually moved a significant by a significant um or by a measure that is large simply because it is large in relation to the Precision that uh that provides the distance between any two points on this so what that means to put it very simply is that time can be measured by the number of CH times I change my mind at level n in my hierarchy divide divided by the number of times I change my mind at

the lower level so now you've got a relational uh you can gauge time by the relative degrees of belief updating you know um the information rate which is the you know um um um confusingly the the the rate of change of the of the length with with time but where time is now uh contextualized by the level above or or by the level below so you have this very relativistic view of time um and you could have different levels you know I would imagine um if you you attend to different levels of your hierarchy you could get a very different kind of time perception indeed we know that attentional set and um um in conditions where um say taking psychedelics or uh in things like Parkinson's disease where where you get um changes in the neurobiological encoding of uncertainty and therefore this fishal information metric you can get distortions and different senses of time um so what I'm saying is there are probably as many different senses of time as there are levels in a hierarchical model and it can only really be measured in a in a relative sense in terms of how many moves do I make at one level in for any given move at another level so um that the what we've done now is if you like um take take temporality off the table to the extent now that we're now just talking about uh movements in an information space or a belief space that has this kind of information geometry these are really fascinating issues uh I think yet to be fully explored um my my friend zaf um um has done probably some of the best work in in um in this area um but you know it is it is to my mind um a really interesting issue because you it really forces you to think about the um what you mean by separation temporal scales and what we what what kind of course grading over time is implicit in this kind of move from one level to uh to the next level does that address your what you were what you were yeah I mean as I was saying this is all really interesting um and again something I hadn't thought of um before in terms of in terms of this in terms of this contextualization process being about moves which implies to me that but there has to be like this sense of time is contingent on these moves that are being done at the different levels of the hierarchy does that process itself not imply some Dynam some temporal Dynamics how can one distinguish a move of 10 relative to a move of five without having segmented moves in some kind of space-time geometry yes and it is exactly that um segmentation that appeals to the information geometry um so yeah how would you actually do that segmentation how how would you measure um the amount of belief updating um in Universal or clock time um at one level of a hierarchy and um this is where the information geometry comes in it's basically the number of different brain states that you have entertained during your movement on this statistical manifold which is also you know I think very nicely articulated in terms of belief updating so you can literally think of this sort of um the space

that would be traced out if I were to plot the activity of all my neurons on every axis of some very high dimensional State space that would be a statistical manifold my belief updating which is a continuous process my gradient flow is literally moving on this manifold the distance I move is the information length that literally scores the degree to which I change my mind it's also mathematically The Information Gain as measured well closely related to the information gain as measured by the K Divergence between my prior and my posterior after I do my belief updating and I'm moving on this manifold and the question question is you know how far have I moved U and can I segment segment that into units and that's basically what the information length you know allows you to do so I I love this picture because if you just think about what kind of flows on this manifold would be prescribed by the principle because the free energy is a functional of the beliefs encoded at each point then the free energy is a landscape on the statistical manifold so we are perpetually falling downhill on this manifold the free energy gradients are themselves and the free energy is itself changing because my beliefs are changing so I've got this lovely itinerant you know model or picture in mind where I've got a manifold it is now where I have gradient flow where I'm flowing on this manifold using a gradient flow on free energy where the free energy now provides a wadington like landscape chasing Minima all the time and of course that chasing reflects the fact that the free energy itself changes as you move around the manifold because your beliefs are changing and the free energy is uh a function of belief and then you introduce this uh this notion which I think is really important the notion that the high level of a deep model provides the context and what would the context look like in terms of this free energy landscape well it means that the landscape would change so you know I've got this gradient flow which is from on a fast time scale just flowing down the landscape down the landscape but at the slow level the landscape is itself now changing all the time so I'm perpetually chasing Minima and moving Minima because the gradients themselves are contextualized and are changing slowly so you get this deep structure to the temporal Dynamics which has this very non-markovian aspect so now we've got this interesting picture where things like you and me that are big where big just means that there are lots of Markoff blankets that constitute my deep generative model that enable this separation of time scales you've got this picture where each level is now providing the landscape for the level below with very very fast changes but of course these are now changing the landscape is itself changing so that now you've got a non-markovian model of a universe that we know is maravian so we

know because we started if you remember right at the beginning with this notion of a random dynamical system that is usually associated with uh things like a lan equation or you know a maravian process um so we're starting from a a maravian um Universe with effectively no memory and now we've found things inside this universe that look as if they found a non-maroven explanation a coar grained sense um sensible and sense making explanation for their sensory Impressions that is quintessentially non- maroven that has this deep temporal structure of the kind you find in language yeah the whole point of say for example um you know hierarchal derish um um process models of natural language right through to the uh current focus on Transformer architectures in large language models they're all about breaking the maravian um they're all about finding non- maravian explanations for what at the end of the day must be uh you know a a Marco a maravian process and you may ask where does that come from well um if we spend most of our time trying to model um input where that input is generated by things like does and we are complex deep things that generate non-markovian actions then in fact our universe now becomes non- maravian simply because we have other things in it that are sufficiently big to have these deep generative models and then you get into language and and communication and uh you know the itinerancy which is I think quite unique to us you know I don't think you'll find this kind of Dynamics if you get too big um and certainly as we've already discussed it it can't exist at a very microscopic scale so even in your head um your macro molecules in the 16th um dendrite in the cal field of your left hippocampus that doesn't do any planning it's too small uh it doesn't have that requisite depth of or nesting of Markoff blankets um uh so small stuff can't do got have this separation of temporal scales and this particular deep itinerancy and sort of um you know Marium breaking like property but interestingly when you get too big um you also can't do it either so you know the motion of the heavenly bodies the uh the moon and the and the Sun and the like they don't plan um and you know it's interesting ask well why not well because they don't have um because a lot effective because they're so big all the random fluctuation in their world have gone away and you now um are left with classical mechanics so you know you could sort of get terministic chaos in end body problems but there wouldn't be a description um that would um be usefully interpreted in terms of planning as inference or active inference in the way that we've been talking about so I think there's a Goldilocks regime you got to be big enough but not too big uh to do this to do this kind of planning wow amazing yeah I mean it's um it's bringing up all of these existential potentially fluffy questions in my mind um which I I I would like to sort of ask at least one which is we could ask why let's take the macro structures

the celestial bodies black hole universe itself we could ask why why in a mechanistic mathematical sense do they not have this kind of non-Marian temporal Horizon so there's a mathematical answer to that which or a physical physical in the sense of being explained by physicists there's an answer to that there isn't necessarily an existential answer to that and I guess I do not expect a concrete answer because I'm sure there isn't one but Millennia has been dedicated to the question of The Human Condition and why maybe that itinerancy is so fundamental defining of our nature which is that we can't seem to kind of sit still we are open and the world is affecting us and we're affecting the world as you said and we're in this constantly Dynamic coupling I guess I would love to just hear intuitive thoughts that you might have on why the universe would unfold in such a way that you do have a goldilock Zone um why why is it the case that things like us have this one wondering openness rather than just being um either in the soup or being firmly rigid forever if you have any thoughts on that um you yeah not well formed thoughts um but I mean it is a fascinating thing to reflect upon um I think the answers that that someone like me would give you would inherit from a sort of um a classical view of um physics as opposed to a Quantum information theoretic view uh so you may get a very different answer if you ask somebody like um um Carlo relli or Steven Wolfram or um you know my friend Chris Fields um but if you're happy with a sort of um 101 physicist's response uh to that um the way that I would look at that is through the lens of the renormalization group which just says that there the um the the things exist at different scales and there is a um you are um looking for explanations of those that that that scale free behavior in terms of laws that are dynamics or in technically um uh that are conserved um over different levels and for the free energy principle that is just the um the Dynamics the gradient flow on the um uh on the free energy defined at each level or each scale um and that leads you to then a view of um the same kind of things sorry the same kind of processes and the same principles applying at each and every Scale of the Universe so that you would at a very very fine time scale and a very very fine um um spatial scale so the quantum scale you would have uh lots of very very fast hot Stones moving around uh where the random fluctuations predominate um and there is very little of this kind of long-term itinerancy um now technically that results from um um some as certain aspect of the way that you can always decompose any Dynamics any gradient flow so I've spoken um about the Dynamics exclusively in terms of gradient flows flowing down um the uh flowing down the uh the free energy landscape say on a cisal manifold being my sort of favorite picture of that um that denies a real another really important aspect of the flow which is the flow orthogonal to that

gradient flow uh known as solenoidal flow or Divergence free flow um so the way that I look that I look at this um through the lens of a sort of classical physicist that that would predicate everything on this random dynamical system or Lan approach um is that you've got a Continuum between the very small and the very big where now we're not talking so much um about sort of the the degree of nesting of Markoff blankets but simply the um the the contribution of random fluctuations to the Dynamics um and the important thing to bear in mind is that there is any Dynamics any flow has this solenoidal part this conservative part and this dissipative part that inherits from the random fluctuations so if you're very very small then all the random fluctuations have not yet been averaged away and the dissipative part predominates over the conservative part the solenoidal part um which means that at the very very small scales you are effectively in the world of um thermodynamics where you're effectively ignoring the uh the solenoidal part um so everything is disappeared you get fluctuation dissipation theorems um you know um uh integral fluctuation theorems you know and then you can drive generalizations of the second law and all of that good stuff um and then as we as we get bigger and bigger and bigger the what the course graining implicit in the um what's known as um an RG operator in the renormalization group sort of I look at grouping and dimension reductions a course graining uh or entails an averaging and as you average the random fluctuations average out to zero so now the the dissipated part disappears and you just left with a solenoidal flow which is just the conservative dynamics of Newtonian mechanics and leanan mechanics and the solenoidal aspect is just a description of like the moon going around the earth and the earth going around the sun it's things that have orbits stable orbits of qu periodic um um orbits that are approaching deterministic Sy they are very very big so the interesting thing happens in the Middle where you've got a mixture of dissipative uh Dynamics and conservative Dynamics what would the conservative Dynamics look like well they would look like this sort of um itinerant um um um oscillation it would look like the same kind of uh classical conserved oscillation that the moon does but it will be much more itinerant it'll look like a a you know um a buyer Rhythm it will look like a life cycle it will look it will have that property that we associate with living movement so you know we talked before about the fact to to um to be autonomous is to move it is to have active States one would also argue that movement in and of itself is not sufficient you know the moon moves um the oceans move um are they biotic in their self organization I think you'd be looking now for a special kind of movement which emphasizes the solonoid a mixture of solenoidal and dissipative uh dis dissipative Dynamics um so sandwiched in between the very

big and the very small is a goldilock regime where indeed you do have this opportunity for the biotic kind of self-organization that characterizes our existence that is um not only um manifest in terms of delicately crafted solenoidal diversion 3 Dynamics but also nested at different time scales so just think about all the different time scales that your um body entertains solenoidal Dynamics um um rotational Dynamics from the fast gamma oscillations again in that hippocampal neuron through to um your um cardiac cycle through to your respiratory cycle through to the your dial Cycles through to your um um slower Cycles right through to your life cycle uh and so on and so forth and then you take it up to any scale you want um but the key thing the point being made here is that not only do you have this ad mixture of disputed gradient flows that underwrite self- evidencing this is in the Context text of this solenoidal flow at multiple scales at nested scales so again we come back to of this deep nesting um so on that view there will be a Goldilocks regime and of course it is only at that in that regime will you ever get systems that start to self- moral and communicate and co-construct their their niches will you get cultural Niche construction will you get the kind of would you get language and will you have conversations like this it's not going to happen um at the level of uh macromolecules or cells and it's not going to happen at the level of heavenly bodies but it will happen at our level um um uh and clearly has happened if I can believe my senses yeah I guess the yes I guess the I mean my followup my my inevitable followup would be why are there these different sizes but I don't I don't want to get into that book it's just Turtles all the way down about why I mean we we can bring it to CL very quickly but that's it it is Turtles all the way down that's I I think a fundamental Insight uh and once you once you commit to that everything starts to make a lot more sense so um you know just in terms of um you know how long will my Markoff blanket last well it will last the you know for a long time at my temporal scale but the um the scale above um will last much longer so I have an environment and a niche that lasts longer than I do and my body lasts longer than anyone's cell and my cell lasts lasts longer than any well intercellular component and you know that component will last longer than any particular macromolecular uh configuration so uh so it is all the way down and you need it at each level you have to have the context being a thing having its own Markoff blanket at its scale in order to um provide the context for the faster coming and going of Markoff blankets at the lower scale but in the same sense because the larger scale inherits from the dynamics of the lower scale there's is a circular caety which people like George Alice would cast in terms of top down and bottom up causation put simply what that means I cannot exist well let's put it another way my hippocampal cell cannot exist

unless my brain is in good shape my brain cannot good shape unless I'm doing good active influence and relating correctly within my family and my consp specifics my cons specifics cannot uh work uh exist unless there is um a biosphere that sustains that kind of Life the biosphere cannot exist and and so on and so forth all the way up and all the way down and every Lev there are things and those things have to have Markoff blankets and at every level they can be construed as doing some kind of Elemental inference or self- evidencing um the the story we're telling though is the special kind of self- evidencing that comes out at a particular scale but the existence of that scale requires exactly the same um I repeat from the perspective of the renormalization group it requires exactly the same Dynamics or LR or functional form of the Dynamics um at you know at every level and you know on this view because the explanatory Target or the free energy principle is basically relational it's trying to self-organization um of a thing that can be individuated um in this context and in terms of what it is contextualizing it's a very myopic um ambition it doesn't describe you know cosmology or um you know the universe uh you're not going to get electromagnetism from the free energy principle um um um and so it doesn't really have to worry about when does this you know whe what is the largest scale what is the smallest scale you can keep on going all the way down and if you had to headed your bets because I guess it is a hedge better bet hedger rather um does it stop does does the is or is it just it just goes on Mark of blankets on Mark of blankets yeah yes mathematically yes yeah there is yeah it's a little bit like you asked me uh where's the um um where's the edge of the Earth when does the Earth stop yeah if if if you're committed to the renormalization group um now it just keeps on going forever um that um and you sort of I'm quite comfortable with that because um you know it emerges from a sort of of um a question which is what is a state so we've been talking gbly about State spaces cisal manifolds and states of Affairs that hidden States and latent States um to have a state space and particularly specifically to have a um a state space that be that can be interpreted or construed as AAL manifold you have to have States but what what where do states come from U and the argument sorry I shouldn't ask you AR the argument is that they are the states of things where the things come from oh we just said it's a partition of States space where did the states come from and if you try to answer that question um then you all you have to do is basically um determine the mapping between um things or particles and states and the answer is quite simple it is it is basically one way of looking at this sort of um or using the appar apparatus of the renormalization group that um the the the state of a thing is the uh state of its Mark of blanket States um and then the um the then then that state itself now has a Marian or a

a a partition into maroff blankets so you now get blankets and blankets um and so that you can resolve the par Parx or the um the Catch 22 you know what's the state of something well what's a thing well a thing is determined by conditional independen among the states you know you can resolve that infinite regress just by saying it is just a recursion that you know States at one level inherit the averages of the markof blanket states of things at the next level but the at every level there are things so you have this um you know you just keep on going down I should say um you know there are people who believe that um there is a smallest scale or it is sufficient I'm not absolutely sure about this but if you ever get the chance in later life if you can talk to uh again people like Carlo relli or Steven wolam um you know from the point of view of quantum luk gravity and from the point of view of the ruad um then the you know there is this notion that there is some irreducible um um smallness scale Beyond which you can't go and then the game is how do you get a continuous world out of this sort of quantum discret um um very very small description of things yes yeah yeah yeah yeah strange Loops makes me think of Hof Stell and strange Loops also makes me think slightly of the um holographic principle what you just mentioned there about projections onto the screen um this this this notion of um reading of the mark of blanket you said something that I said we had three things that I wanted to talk about um and we covered temporality all beit over a long temporal scale wanted to very quickly ask you mentioned um in passing um about celestial bodies being nearly deterministic and my ears pricked up um I on an internalist model of the free energy principle I sense that there's actually a kind of Licensing for free will because we have this internal model and we act upon the world as to confirm the model that we we wish to we wish to observe on an externalist perspective where we really are no different from another physical entity except for the fact that we have the possibility of moving things and and acting to to self-evidence there isn't so much space for free will but my ears picked up when you spoke about agency and when you spoke about the celes your body's being nearly deterministic because it makes me think it seems to imply at least uh huris that being open to the environment being an itinerant being licenses agents agency but I want to make sure this is true metaphysical agency in the sense of not hard determinism um or whether this is agency in the sense of self-conception as a free agent because those are two different things um it's we won't be denying that we perceive ourselves to be free I guess the fun the more fundamental question there is whether that's an illusion or not and so I was wondering whether free energy principle has anything to say about a kind of libertarian versus deterministic perspective on Free Will um well I mean you could certainly use the the the um the free energy principle to

tell stories that speak to those arguments um and I think you picked up on the some you know some of the key parts of those stories um so you know deterministic chaos that you get with nobody problems when thinking about so massive bodies dancing around each other um you know there is still chaos in mix in the mix um so there's an unpredictability and sensitivity to initial conditions in in the good oldfashioned way of um non- dynamical system System Theory of A deterministic Sort um but once you get to um you know things like you and me um you're really in the world of stochastic chaos so the stochastic bit inheriting from the uh the random fluctuations where you've got um two kinds of um if you like or two aspects of uncertainty and implicitly then in modeling that uncertainty um um uh the opportunity to talk about sort of Free Will and selection um one interesting twist here is that the um if you actually want to apply the free energy Principle as a principle of least action to things like you and me um what you have to um um imagine is that we are deterministic things in a stastic and chaotic World um so in making sense of things we only do so at a particular scale we're talking before about the averaging away when I'm sufficiently big to average away my random fluctuations under the free energy principle that literally means you have to take the average of lots of neurons so the kind of sense making the kind of uh dynamics that you would simulate under the free energy principle applies to and only to The Ensemble Behavior where you can average away any of the stochastic aspects so you will not get you will not be able to deduce evidence or um um solicit or um find evidence for the freey principle by looking at single neuron single synaptic it has to be at the The Ensemble level where the the um well there are all sorts of arguments where that has to be the case but um from from our point of view it's when the random fluctuations are averaged away um so I'm just qualifying what I'm saying is I'm not saying that the brain is stochastic or shows stochastic chaos what I'm saying is that the brain has to provide an apt explanation or approximation um to explain stochastic chaos um and crucially um the um stochastic chaos associated with our own behavior um with the consequences of our own actions so you got a deterministic system trying to now Model A non-deterministic stochastic chaotic system um and you know in so doing well that's easy to do because of course we're we're talking about um that modeling being in terms of belief or probability distribution so that that's Absol abely no problem um so um on that view I think you could easily motivate um Free Will of a certain kind um in the sense that you when are when you're planning you have effectively to select amongst different models of the future technically this we Bas in model selection and selection is an act and so again we have a different kind of agency here but it has it's a kind of agent inherent in the selection amongst

competing plans or policies narratives paths to into the future um that I think has all the space to um um say that this and it's you doing the choosing remember you can't there's nothing on the outside that can influence directly that choice on the inside so from your point of view you certainly have free will would it look like you have Free Will from the outside um um I think that's a more vexed question uh of course you'll never know but but uh I think you'd you probably You' be in the game of of of a game practically how I spend most of my um um day job is inferring to what extent is this artifact or this person um doing good um planning as inference good planning um you know for example if somebody's in a or in or asleep or has a psychiatric disorder um you know you you need to know to what extent they are um and in what way they are self-modeling um so you know um this is a slightly sort of clinical and Abstract example but illustrates a point that you will never know whether something has free will all you can do is infer that they are behaving as if they had a generative model that entailed a self model and and thereby a model of the consequences of action and therefore as if they were a true agent and possibly they may know they're a true agent they may not you know they may be again um you know that would require another level of of the Markoff blanket is that what you had in mind yes I mean I think um I guess the the the kind of Hardline determinist response would always be pushing you to say whether that organism or particle could have done otherwise so it may have this reflexive quality of okay I'm doing basing model selection to my sense of self and we will come to Consciousness because I think that's the kind of Crux of this issue is why is any of this happening online anyway but it it might have this self um reflexive notion of being a free agent but the Hardline determinist from the kind of external position of physics might always say well could that bundle of atoms ever have done otherwise in terms of its basy and model selection but it's it's again it might just be one these sort of Loops that just keep spinning um yeah it it's it's all um fascinating I had a question that I've just popped out of my head when I was talking about Free Will which was um I will come back to it I it will come back to me um Consciousness I know it's I don't know if it's being considered I don't know if we can call it a red herring yet in um the academic world because it's become a very hot topic um and people are getting very heated about it and there's no need to bring up particular examples of when people have got heated about it but people are very invested in their theories and I guess rightly so it it seems to be this one thing that people can't seem to couch within their theories of everything if they have one do you let's starting with very simple question do you consider it to be a real problem do you take something like the hard problem seriously or for you is it more of a meta question or can it or

does a deflationary account just do the job in terms of the hard problem um it really depends who I'm talking to as my friend Jeff Beck would say it depends which pants I'm wearing so I think I think very much it's it's a question of um a conversation and you have to you have to infer how much investment the person you're talking about has in this issue and what position they're taking um and I mean that you know just in terms of being polite and engaging you know with Integrity in scientific uh debate or indeed philosophical debate but I also mean it uh in a slightly more fundamental way that Consciousness is is an illusion um and but I think Illusions are fantastic they're literally fantasies and I think that's what makes our brains fantastic organs I think they're hypotheses I think that the the illusion is um the thing that we uh we test it is the thing that generates the predictions in predicted processing so I think Consciousness is another illusion um very much along the lines that one reading of Andy Clark's basing qualia paper um um that even qualitative experience quaia are just Illusions they're just things they're just constructs we bring to the table including selfhood um they're just hypothesis illusions that are particularly apt at um explaining everything that we sense and making good sense of it as as a parsimonious H hypothesis Consciousness itself I think is exactly that because it only applies the hypothesis is that um is this thing conscious or not only applies because there are other things in my universe that I'm trying have to model like you so it's the question is not whether I am conscious the question is whether you are conscious and of course we've already said that is unanswerable because you have a markov blanket so it's an unanswerable question but crucially it's a question about the things are observe that I'm trying to make sense of um it only becomes I think um only in in the spirit of the meta hard problem or The Meta problem um it only becomes um um vexed when you make the mistake of asking am I conscious I don't think that's what the question is there that's the illusion is not for me it's for you uh the illusion is the Consciousness is a construct a hypothesis and a very plausible and all the evidence that I have at hand would suggest that you are conscious and indeed literally because there is that evidence um it is there in the sense of an Evidence maximizing illusion hypothesis or component of my generative model um so I I go through that just to make it clear that that you the answer depends upon who you're talking to but also who you are trying to um ascribe Consciousness to so if I was talking to um um well if I was talking to um people like yakob ho um um you the first thing he'd say is that the free energy principle is not a theory of Consciousness and in fact he likes saying that every every at every opportunity so so there's nothing there's nothing um there's nothing in the free energy principle that has anything explicit to say about explicit to say about Consciousness on the other hand people

have used it in different ways to uh address the hard problem um uh you could take the view that Andy Clark takes that there isn't a hard problem and perhaps even with David Chalmers' work recently on The Meta problem The Meta hard problem that's probably a more interesting problem um or you could grasp the nettle and start to talk about it if you were talking to um my colleagues Chris F and um um and um Adam Saffron and Maxwell Ramstead and listen but um certainly um so I think inspired uh by something that you mentioned earlier on which was the holographic principle in um um in Quantum Information Theory where the holographic screen now um plays the role of the Markoff blanket that separates the inside and the outside in terms of bulk reading and writing to um the um the um the holographic screen um if you're talking to him um then he would start to talk about these minimal screens uh these minimal Markoff blankets um and develop quite a quite a sort of nice argument that if you can't reduce in the sense that we're talking about nested Markoff blankets inside a Markoff blanket if there are no more reductions at hand then the only way that the internal states of that irreducible Markoff blanket can know about their own existence is by acting upon lower levels and therefore there's some um there's something unique there there exists a unique Markoff blanket that that that you um maybe um uh have the attribute of of Consciousness um if you're talking to somebody like Van Vleet um um he would you know tackle this in terms of um trying to resolve uh a dualistic position by offering a dual aspect monism um of a Maravian Sort and that rests very heavily upon this information geometry so if you remember before we're talking about sort of a state space equipped with free energy functionals um on it um um that um plays the role of a causal manifold but but there are two causal manifolds that are jointly situated in this in the same state space of say neural Activity one is that for any given uh point there is a probability of being there from a point of view of neuronal activity and the thermodynamics so this would be a thermodynamic um kind of information geometry and then the other one is what the free energy principle uses which is the encoding using that point to encode beliefs about external States so you got two um information geometries belief structures if you like um that supervene on exactly the same um material um process one of which is pertains to probability distributions or beliefs about the internal States and the other one is probability distributions or beliefs encoded by the internal States about the external States so you could construe one as being the mental and one as being the material and then you've got this dual aspect monism where you're now asking questions about well what are the lawful relationships between the thermodynamic free energy and the um the um the variational free energy that pertains to beliefs about things so you've

got now a mechanics that talks about beliefs and that may well be one way of repairing that sort of a dualistic approach um there's another another perspective which you would get if you were talking talking to talking uh to people like um Julio tanon and um SEL um well my colleagues involved in a one of the Templeton Foundation absera research uh um collaborations um where you're trying to find Bright lines between the free energy principle and integr information Theory um um which is not necessarily the best thing to do because a lot of integrated information theory is exactly consilient with the free energy principle however if you are going to get funding uh by uh trying to tease them apart then the thing that distinguishes those approaches to um uh Consciousness is really the the active part that we were talking about so uh boiling this down in a way that some like I repeat yakob Hoy might might articulate this is It's the difference between seeing and looking you know to be conscious I have to look it's not enough to see um it's um you know to be conscious I have to listen it's not enough to hear uh so you're putting action into games there active sensing this this sort of closing the the loop between the inside and the outside may be um the thing that um is necessary for Consciousness and simply because you're saying that Consciousness is an attribute of agents and agents have to act what do they act on their Sensations what does that mean they they do active sensing so to be an agent is to be an active sensor yes active inference um and you if you're saying that things that aren't agents can be conscious then you wouldn't worry about that but if you are committed to a deep um to the notion that to be conscious um of to have a minimal selfhood um then you have to have the notion of self if you have to have the notion of self then you have to be an agent if you have to be an agent you have to um um have generative models about the consequences of your agency your actions uh and that has to have the temp then we go through all of those nice links that we were talking about before with temporality and and you know and the like and indeed the hierarch structure takes us back to the Chris Fields um irreducible um uh holographic screen or or um or Markoff blanket so that would be the way that I would um converse with that crowd uh uh I run out yes I run out of people that I might talk about it so I have to find out what you like and then i' I'd tell a story that that you like or let you tell a story that uh that you like well I guess my question to you would be can you imagine a world of self evidencing can you imagine okay let's put it another way can you imagine a perfect replica world like ours where the lights are turned off entirely there is no consciousness or is it do you imagine it's a necessary uh consequence of the way that creatures are like like us are structured that to do deep temporal planning to do planning as inference what whatever it is or self hooding we need

Consciousness could there have been another way that the Universe would have unfolded without us being online um I suspect not um sorry I I'm torn between um um been taken by you down sort of your philosophical zombie uh routes versus um another um story that comes out of the free energy principle about Consciousness which is um the um the the importance of self- modeling in relation to other modeling and the fact that we are actually living in a world um of things like us so and one story I didn't cover which is a story you would tell if you were talking to um my friend Chris fth or people who are much more interested in the sort of social neuroscience and uh or say attachment Theory um and you know developmental issues um that are induced by a sort of evolutionary psychology view or Niche construction for example all of these views of Consciousness means that Consciousness is just there um which is where I started to infer um the state the disposition the intentions State of Mind of somebody else um why and how can you do that well you can only do that if that thing is sufficiently like you to be able to use your generative model um as OB say your physical actions as a model of their physical actions to infer their intentions and the like um and when one takes that to its limit then um often what the real problem is is not in terms of inferring the content of some communication or some uh exchange of of um semiotic cues it really is inference about agency again it's did you do that or did I do that so that ambiguity now requires you to have a model of self versus other so it's a much more relational um um notion of self so you know sometimes I think people talk about minimal selfhood um as something which is monolithic I don't think that's necessary as part of a gen model it certainly would be important to have um different um different kinds of self you know am I embarrassed am I in love you know as we were talking about in you know in the um tmbb meeting a few days ago that would be you expressive and very necessary part of generative model where I have to contextualize my own emotional and pro-social and autonomic and moic behaviors um but there's no one self it's a self you know self like this like this and like this a more fundamental relationship relational aspect of selfhood is in relation to otherhood you know mother uh you know versus me and am I mom or is is mom a thing am I a thing I'm the same thing as Mom you going to S clan um uh sort of arguments there so put simply what that means is it may well be only in universes or worlds that are populated by ensembles of conspecifics that are sufficiently similar would you need to self Ood simply to distinguish self from other in a universe where there's only me I wouldn't need selfhood so it would not be it would not pay its way in terms of model complexity or model evidence to have the notion of me being a self if there's only me it only pays its way if I now have to entertain the hypothesis

that either I cause that or you cause that so the question now is is it inevitable that you get a universe with lots of similar things emerging all disambiguating themselves or attributing agency to themselves in the gold uh in the goldilock zone I think it probably is I think that would be I think if you simulated sufficiently accurately artificial life I think you would you would get the emergence of um populations of similar things and they would Niche construct um and they would be thereby Drive their own um sort of Nest increasingly nested and um structurally complicated um or at least sparsely um structured um uh organizations in terms of their Markoff blankets um and that this would um this at some point would inevitably invoke um modeling of others and therefore almost by default a a a a minimal notion of of of selfhood I think that would be emergent you but it would require being in the Goldilocks regime for a sufficient amount of time for this kind of Niche construction uh cultural Niche construction to actually play out so you know you'd have to have this very slow time scale providing the right kind of context for this this kind of thing to uh thing to emerge yeah yeah I think my well I have some I think I have some sympathies to the to the argument that Charmers would have laid out in or did lay out in 95 in his original hard problem paper which is that I think I can still ask the further question which is can systems like me make the functional distinction between myself and others without there being a sense of what it's like to be me can that just be an offlin which is done purely in some algor rhythmic manner which need not involve some qualia of my selfness so yeah there there's a point that there's an element in me where I don't want to say that that in a sense resolves a hard problem because at least I can hypothetically Envision that being done offline um I wonder whether you've given much thought to Mark SS's idea which is that Consciousness might be rooted in affect because this seems to me to be a bit more of a self-confirming idea which is that feelings must be felt now I think there's still an open question there which is affect arises at the point of uncertainty uh and drives us to resolve uncertainty and when we do so we feel good about ourselves now again I still think when I was reading his book I still had this open question why would that have to why would why would affect even emerge in that place why wouldn't we just have this drive to resolve that uncertainty without there being the onliners so at least for me although I may be wrong and I'm very happy to be wrong there is still no simple parsimonious explanation which can get around this um this point that Charmers makes which is you can give me the function but you still you can't give me the explanation power why it needs to be online um and I wonder if there ever will be some mechanism which is sufficiently which isn't just purely tautological or invokes consciousness as some fundamental reality which makes

me which makes me wonder whether you've heard of Donald Hoffman whether you're familiar with his work yes because I sense that in some ways it has some interesting implications maybe for the free energy principle in the sense of space and time not being fundamental units of reality we've been speaking a little bit about how um where do space and time come from how much do space and time overlap in the sense of generating um like we need expansive creatures like us with embedded Markoff blankets inevitably have a sense of temporal modeling and there's these interesting interactions does the idea of space and time being what he would say applications on a desktop does that have any real do you sense has any real ontological implications for free energy principle or is it a red herring um well you you've introduced two important things I forgot to mention Mark SS there another friend so to Mark SS I would say it's all about this irreducible Markoff blanket that acts upon the rest of the brain where is that action it's going to be in the neuromodulatory systems set the Precision the Precision is exactly that Fisher metric which tells you about how much you're updating your beliefs it is the way that you do mental action it is as you intimated um feeling in the sense that it is quintessentially balanced by increasing or decreasing the Precision or uncertainty so um you know that's how I would tie in Mark's um formulation to which I think is internally consistent with um say Chris Field's formulation or um Van's formulation so Mark SS focuses you know on those I think a key aspect of um of active inference which is uh if you're a psychologist it would be attention it's basically acting in a way on the inside to um get the an estimate and deploy the right confidence or the right Precision in um um mathematically the curvature the of of of the um the free energy in and of itself um getting that right um and um you know that has a lovely you know connection to attention means that you can't divorce um Consciousness from attention you can't divorce it from agency in the sense of mental action or internal action you now have a deep connection to um Thomas Metzinger's um distinction between you know um opaque and phenomenal transparency um and uh and the way that Mark would tell the story would be that this is just feeling and feeling is at the heart of of any qualitative experience um so you yeah thank you for the for reminding me he was one person I forgot to to cover um the uh the second thing um you know the the space and time being Illusions um I mean yeah I sort of start off there so I'm not quite sure that it uh so if you're saying that Donald's um and I should say he's a friend I think I've never met him but I think he's a friend of Chris Field and Mike Levin so you know there's a small world here um but if you're just saying that um space and time are um Illusions in the sense that we were talking about Illusions as being fantastic and the thing that makes us into fantastic little beings

and have beings and um uh and our brain into a fantastic organ yeah absolutely I mean you know the question for me is why on Earth would you um what what what must be the case for us to have this illusion of SpaceTime and that we are living in a metric space and it's practically very important I mean a lot of people spend a lot of time in machine learning embedding things in metric spaces um and sometimes it's very difficult to see what licenses then to do that uh you can certainly see how it would work in terms of projective geometries in metric spaces um but the Deep question is why you know you know I can't um I can't imagine a um you know an amoeba really having having a notion of of uh space um I may be wrong um and so knows um um you know if you listen to people like um um Steven walram um you know the notion of SpaceTime um itself May the illusion of SpaceTime itself may be just an artifact the fact that our sensory organs respond very very slowly in relation to the speed of light so if light traveled very very slowly or imagine a world in which um um we saw with sound and sound was very very very slow um would that would that be would that give you the illusion of space um you and distance at an action of the kind that you we sort of implicitly assume when we're talking about sort of visual objects so I think he has this nice notion that you know in this kind of World um with these kinds of sense organs um you know when something moves from here to here is it the same thing anymore because that's what you're saying when you have a metric space in which objects move um you know some certain symmetries or invariances are inherit from the notion of movement in a metric space but that may well just be be be be a fantasy we got gone off qualitative experience in the hard problem yeah well no I by it all it all ties back it's a Beau it's it's it's a very good question which is why would this why would space and time be a an the interface for us I guess that is the Deep question um and yet it appears that that at least it it is something at least in conscious awareness I did remember my question and it actually feeds into the last couple of questions I will have which is you mentioned um that you spend a lot of your time inferring in the technical and Layman sense whether other people have I think you say good generative models there's an interesting question here I think which which I may have had intuitively when I started reading about active inference which is is there any such thing as good self-evidencing in the sense that I may so so a lot of as of course you know as you wrote a lot of the papers a lot of psychiatric disorders and psychological disorders have been rooted in um disruptions to Precision waiting mechanisms for example can we consider are the words good bad optimal suboptimal are the these useful terms in this context or is it that that individual has a set of priors which are different from you and I and everyone else and we all actually have a

sense of we all have kind of different prior so all self-evidencing in unique ways and yet philosophical vagueness allows us to Clump us together into a group is there such a thing as what it is to be a good self-evidence as a human with beyond beyond just a trivial point that we're not just dissipating into the the Heat right yeah that's an excellent question um so I I think the answer to that is yes um but as you correctly um pointed out it should be qualified by the um the complete class theorem So you you're clearly referring to the complete classroom so it might be just wise to um just Define what that means and the implications for discussions of good and bad um um generative models um so the complete class serum way i' like to summarize it is that for any given pair of behaviors and loss functions there exists some prior that render your behavior Baye optimal which means that there is always a description of every Behavior as base optimal under some prize so in that sense you can't be good or bad um where the thing is basically your decisions because this inherits on based in decision Theory so it's it's not quite as full uh active inference in accommodating um information game but um within the within the um confines of phasing decision Theory you can't be um you can't be good or bad you just have different priors so you my favorite example is that you know if we all went to live on um on mars or a much heavier planet people with Parkinson's disease will probably be much better have better prize you subpersonal prize in terms of that don't be deficits because yeah they move more slowly and sort of more cautiously so yeah I think from the point of view the complete class I think it's important to bear that in mind you know there are just different prize but I I do think though you can sort of just say well hang on a second what about whether a good and bad prize and I think then we get into this again this hierarchical story because of course as soon as you have a hierarch generative model there are no prize you everything's an empirical Prime everything can be cont realize everything is conditioned upon everything else including your prize and then you're just back into the um into the game of self-evidencing what is good self-evidencing it's achieving a higher marginal likelihood or model evidence um how would you do that by finding the right prize how would you do that by learning how would you do that by changing your model basic model selection or structure learning um how would you do that in evolutionary scale you do basic model selection with natural selection so yeah um on that view the log evidence as bounded by the free energy um is just adaptive Fitness is how well your prize are adaptive to this world you are trying to explain so I think yes absolutely you can you can you can be a good self-evidenc you can be a bad self-evidenc if you're a natural selection um that matters because if you're a bad self-evidenc that means you won't be around um in the um in the Next

Generation so the marginal likelihood of you existing at an even time scale is low so the your evidence is also the probability that you will be found exist you you will exist so it's quite important to get to be a good self-evidencing and because that's your marginal likelihood of existing uh and I've now slipped in another synonym for um free energy Beyond um prediction weighted prediction errors or Beyond on surprise or surprise or self-information Beyond log evidence uh it's also statistics called The Log marginal likelihood it's the the likelihood of your sensory exchanges with your world your econiche having marginalized out or averaged away all the unknowns which in this context are all the states out there you know the hidden causes or the sorry the external states which you model in terms of hidden uh in terms of hidden causes um so yeah I think it's very important to be good and bad at evidencing the measure of goodness of Badness is just the marginal likelihood of the modal evidence and the free energy there is nothing else um uh if you're saying does that speak to some creatures that will be able to actually assess how good they are self-evidencing then the argument say absolutely um people like Matti Joffe um looked at this specifically from the point of view of emotion and developed a whole taxonomy of emotion in terms of um rates of change of free energy and you get into stories of slope chasing uh um looking for gradients and gradients and gradients of uh self-evidencing on the measurement of the goodness of self-evidencing which is uh which is which is the free energy um more refined stories more recent stories um would basically be looking at certain kinds of uncertainty um being encoded explicitly in your generative model as underlying affect uh particularly the the your certainty about what you're going to do next for example then we get back to um the story that Mark Susskind would like to tell which is um the Precision of your beliefs is the most you know the most important attribute that defines um you know its vigilance and that your your feeling at this sort of minimal effective level yeah yeah the complete class firm is an interesting one because I think it leads people it leads one into a potent these potential strange conclusions um it makes me think of animals that commit suicide oddly so um I think sort of AIDS might explode themselves in the context of a predator It ultimately comes back down to your model evidence I guess so I am the type of creature that will commit suicide if I'm in the presence of a of a ladybug of course but then those kinds of examples I sense for me show that the words good or bad I'm just pushing a little bit back just intuitively the words good or bad in in in in the form of uh continued existence slightly put the cart before the horse because who said that that's the imperative so if I existed in the world where my everyone's imperative was to no longer exist you know know God forbid then that then I would be doing bad self-evidencing

on that standard if I didn't if I continued to persist so is there a point at which even the yeah like do you see how well do you see that argument holding any water um or is it just now we're just dissolving into semantics well no no I think I think it's an interesting thing to to contemplate uh you you you you're you're basically I think undermining um that particular argument by uh um proof abductory absurdum um so if you think you are the kind of thing that will uh self-destruct or aviate or commit suicide um then uh you can't exist because you killed yourself right yeah so I will be extremely surprised um if there were any um uh common natural um um subhuman artifacts that kill themselves I didn't know about the aphid example very I mean obviously Lemmings is is the sort of popular culture one but um um you I think they'll be extremely rare and you and then you have to understand the um selection at an evolutionary or transgenerational level um um in order to apply the free energy principle um but in you know I think you're absolutely right I'm very mindful that a couple of weeks ago I was asked to review a paper applying Act of inference to suicide and it was a very Artful paper um and uh I do remember thinking I'm not sure whether the point was made implicitly or explicitly in the paper but you know it's it made me think in exactly the way that you were you were um talking about um you know the imperatives to survive and and are they the right way to think about the causes of our behavior um I I would say absolutely not not this is not about um wanting to survive this is simply wanting to avoid uncharacteristic Sensations and then we put a post talk hypothesis oh that's because I want to survive then right but that's a fantasy that's another illusion uh so you know we do not know what it is like to be dead so yes it cannot be well it can be surprising but it certainly can't be the kind of surprise that you'd register in terms of a free energy so it's not that we're frightened of being dead we're frightened um we are going to avoid bya these pragmatic constraints um states that we've have inherited that um for portend um cessation and dissipation and death death in in all itself is not at all frightening we go to sleep every every um suspension of got to go to sleep every every night um so you know I I I would I would um sorry yeah and the final point I think is I think what you were doing though you were probably um creating a slightly vexed and self-defeating argument by saying uh by assuming that um we can understand things in terms of imperatives the whole point of the free enery principle is that there aren't any imperatives this is where it starts as things that exist must behave like this and part of the behaving like this means that you can interpret them as optimizing something and complying with imperatives but that's just an interpretation that you'll bring to the table when you apply it to something else if you then apply it to yourself then you know then things might get interesting you might have hard

problems but there are no imperatives you know in fact even minimizing free energy is not an imperative you can have the free energy principle applying to um things that have very dissip uh dissipated attracting sets um that have very very um High entropy um all it's saying is that there has to be a certain uh gradient flow that count as the random fluctuations then you ask yourself well hang on a second so why does it look as if we're always trying to attain some very precise homeostasis um and the free energy principal would turn that on its head it would say well look okay in some universes there may be very precise things um and these precise things what would they look like oh it would look as if they were trying that they had um aspirations to EV evincing a very precise homeostasis and indeed allostasis so you get this notion of another kind of natural thing which are precise things um so if precise things exist it would look as if they um they are um compelled to comply to very specific imperatives but from The View it's not saying they exist because they comply with the imperative it's saying existence implies it looks as if they have this imperative right now one final point you know I'm I'm aware that we' we've now approaching three hours well final point though I you introduced the word optimization I think that was um the right a right and important um um thing to introduce because you know it's only when you bring free energy into the game as a bound on marginal likelihood or self the negative log marginal self-information or surprise or surprisal do you convert an existential description into an optimization problem so that's exactly the move that Richard fan made when he introduced variation free energy he had this impossible marginalization problem that to actually evaluate the marginal likelihood of paths so exactly the same kind of problem that we're talking about in terms of uh what my prior over policies and plans and paths into the future um you can't you can't evaluate that exhaustively it's analytically and practically physically not realizable so the marginalization problem the true inference problem if you like uh the true existential uh problem if there is one um cannot be solved but what you can do is introduce this and turn it into an optimization problem so just by using free energy um or just by appealing to this Tey that comes from um predicate everything on this evidence lower bound you are actually saying that it looks as if creatures that exist are optimizing something what does it look what are they optimizing it looks as if they're optimizing a bound on their um marginal likelihood or their adaptive Fitness um what does that look like well it looks exactly like basing inference uh or if you're talking to somebody machine learning it would look as if you're um optimizing your negative variation for Energy is known as an elbow so this is a evidence lower bound it is exactly objective function used in um sort of high-end neural networks like variation Auto encoders um so you

know I think that the distinction between um just being and optimizing is quite crucial and you're absolutely right that once you make the move um of um framing things in terms of variational bounds using variational calculus or variational Bays um then you have actually committed to an optimization narrative but you didn't have to just to exist you you you could uh yes yeah yeah I I was gonna have this funky philosophical argument about needing to exist to predict but I won't go there I won't go there um because we are going to wrap up all of this philosophical mathematical thoughts has and Consciousness thoughts has brought to my mind Thomas Nagel um Thomas Nagel has this idea that death is bad because life is inherently good and we've been talking about life and being and what it is to be and what one needs to do to be and persist to be if we're going to wrap everything up is there a sense in which that kind of related to what we were just talking about in terms of imperatives of course there is no imperative to be in any sense but you said it was an interesting thing you said there was it looks like we are all being anyway right even if we didn't have to it looks like we are there's no God granted plan for us to be but we seem to be doing it anyway does any of nagle's romantic philosophical idea that life is fundamentally good being is good does any of that resonate with you um yes it does because it's it's it's you know it's nice isn't it and I you'd probably you you'd probably sort of license that with with um phrases of the kind that you know to exist is to be curious um so this speaks to this sort of um you know the ability to respond to um epistemic affordances and certainly you know to exist as curious creatures um is obviously the good way to exist in you in any given world and it's you and would be mandated um I can't resist saying though of course this only really applies to you it doesn't apply to me so so you are good because you are curious and you exist and I wish I could say the same for me but I can't strictly speaking but you could if you wanted to I could I could abstract to you in some degree I could infer that there is a you great um and then I guess we can wrap it up there with one more question we've just spoken about life being good I'll flip the question I don't think I've ever heard you talk about death although we briefly touched upon it um I won't make it too personal but if life is kind of good in some sense because of the concomitant parts of living as we do as I don't know if it's good to be a stone but it's nice to be me having these conversations would that make death bad is that is there is there a is you said we fear dying because we fear in an affective sense um that which we don't expect to the states we don't expect to embody and in some sense we don't expect to embody death because it doesn't fit in with our priors but beyond that free energy principle perspective is there a broader position in which just returning back to the fabric of the universe is bad is there

something to say about re-entering that Mark of blanket does do my mark of blankets dissipate um what what does that process look like and and does that strike fear in you or strike awe um well more or but I don't want to overstate it um it is it is beautiful um from a from a p mathematical point of view um so I I have a very sanguine view of this and we've already said you know you've never been dead I've never been dead so no one can tell me it's an unpleasant experience um so I I'm not particularly I I I'm not going to um avoid being dead I certainly think all the all the paths to being dead are probably very surprising and unpleasant so I'm going to avoid those death in and of itself I don't think is uh is sort of um quintessentially bad um in the same way that um Nal being curious and being engaged with the world is a good is it is good while you exist um um I think that's perfectly okay but I think death is um or the life cycle let's call it a life cycle that it's slightly less um um imbued with um morbidity um the life cycle is quite important because um if you just look at um if you just take this um multiscale um um view of um self-organization cast in terms of this basic mechanics um that you know for example natural selection being based in model selection um then um if you put that together with what we're talking about before about um ecosystems and Niche construction constantly changing the lived world then what you need to do is to do constant Basin model selection um in order to get the right structures very much in the spirit that we talking about before where there are good priors and there are bad priors in the sense of the marginal likelihood or adaptive Fitness to learn earn those good prize you sometimes have to start from scratch again um with a different structure and of course this is exactly what pay in model selection does via structure learning at an evolutionary time scale will entail death so that picture basically says that we have to keep refreshing sort of sometimes cast in terms of Basal forgetting so we the fact that we die are Basal is is could be construed as a basal optimal forgetting where we are the environment's memory of what is good for this Eco dis but the environment or least natural selection knows that the environment is changing so what was good last generation will not necessarily be good for the subsequent generation so it has to forget in the same way that you learn by forgetting you have insights by dispensing with redundant information so in a sense death for in that life cycle is an essential part of um uh of the self-evidencing of the species so it may be bad for you I suppose you know in a nesque sense in that it's denying the opportunity to be curious and engaged with the lived world but it's certainly good for your children and your children's children right right right I'm a I'm a for warning for them that's good I'm glad to I'm glad to be that person in that game and a kind of cheery note on which to end um will be a slightly morbid question Carl this was an absolute

pleasure um despite the occasional Wi-Fi hiccup I think it's um it's gone you know I've been I'm incredibly happy with it thank you so much for being our first guest um and answering my sweet of sometimes bizarre questions it was a real Joy on my part right well it was been a real pleasure talking to you your questions were excellent I have no idea how long you're going to have to spend editing all this I I thank you again for that opportunity to uh you know to talk about these wonderful issues and hopefully we'll do it again excellent thank you go